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****Project Title**:** Forecasting Geomagnetic Storm Intensity with Multi-Stage ML Models

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Abstract

Geomagnetic storms can disrupt satellite operations, radio communications, navigation, and power systems. This project builds a machine learning pipeline that predicts the geomagnetic storm index (Dst) using solar wind measurements and derived features. The core model is a bidirectional LSTM with attention, trained on an hourly-resampled solar wind dataset spanning multiple decades. The model demonstrates strong predictive performance on 2023, achieving RMSE near 5 nT and correlation above 0.95 on a full-year evaluation window.

To enable operational forecasting, the project includes a solar wind forecaster (multi-output LSTM) that predicts key wind fields 6 hours ahead. This allows Dst predictions even when future solar wind inputs are not available from instruments. The system is designed for fast inference, reproducible training, and deployment in a web dashboard. We report training methodology, model architecture, evaluation metrics, and limitations, and provide a practical path to real-time deployment.

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1. Introduction

Space weather forecasting is a critical capability for both science and infrastructure reliability. One of the most widely used geomagnetic activity indicators is the Dst index, which summarizes the global intensity of the ring current and storm strength. Accurate Dst prediction is challenging because the system is driven by complex, nonlinear interactions between solar wind, interplanetary magnetic field (IMF), and the magnetosphere.

This project aims to deliver a robust end-to-end forecasting pipeline that provides:

- Near-term Dst prediction from measured solar wind and IMF
- A pathway to multi-hour prediction when solar wind inputs are not available in advance
- A model suitable for deployment with a web-based dashboard

The system uses long historical records of solar wind and geomagnetic indices, applies careful preprocessing (hourly aggregation and gap handling), and trains a sequence model that can capture temporal structure. The same framework can be extended to predict storm risk probability and to produce early warnings for operational use.

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2. Problem Statement

We aim to learn a function that predicts the geomagnetic response (Dst) given historical solar wind and IMF observations:

- Inputs: solar wind speed, density, magnetic field components, and derived plasma parameters
- Output: Dst index at a future horizon (1 hour ahead for the main model)

Formally, for a time window of length T hours and horizon H hours, we map:

$X(t-T+1 \dots t) \rightarrow \text{Dst}(t+H)$

We also build a complementary model that forecasts the solar wind fields themselves. That model enables multi-stage prediction for years where future solar wind input is not yet observed.

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3. Data and Feature Engineering

3.1 Input Data

The dataset used in this project contains decades of solar wind and IMF measurements, organized as time series with minute-level resolution. The training pipeline resamples to hourly averages, which balances temporal resolution and noise reduction.

Key characteristics:

- Multi-decade coverage (1995 through 2025 in the processed file)
- Hourly resampled features
- Geomagnetic index labels for Dst

3.2 Feature Set

For Dst prediction, we use a wide set of physical variables, including:

- Magnetic field magnitude and components (B , B_x , B_y , B_z in GSE and GSM)
- Solar wind velocity components (V_x , V_y , V_z)
- Proton density, temperature, flow pressure
- Derived plasma quantities (electric field, plasma beta, Alfvén Mach, magnetosonic Mach)
- Position parameters and time markers (day of year, hour, etc.)

This broad feature set is intended to capture both the IMF coupling mechanisms (e.g., B_z) and global solar wind driving conditions (e.g., dynamic pressure and velocity).

4. Preprocessing and Data Quality

4.1 Hourly Aggregation

The model uses hourly mean values to reduce noise and align with the Dst cadence. For each feature, we compute a 1-hour mean.

4.2 Missing Data

Two strategies are supported:

1. Interpolation (cubic spline, limit 24 hours)
 - Fills gaps with smooth estimates
 - Avoids extreme artifacts by limiting long gaps
2. Drop missing rows
 - Strictly uses only fully observed hours
 - Reduces sample size but avoids imputed values

The 2023 full-year report used the drop-missing strategy, yielding strong evaluation results. For 2025 prediction, missing features were filled with training means to allow forward inference (documented in the report).

5. Model Architecture: Dst LSTM with Attention

5.1 Core Architecture

The Dst model is a bidirectional LSTM with attention and normalization layers:

- BiLSTM (150 units) -> Attention -> Dropout -> LayerNorm
- BiLSTM (170 units) -> Attention -> Dropout -> LayerNorm
- GlobalAveragePooling
- Dense output (1 value)

This architecture captures both forward and backward temporal context, while the attention mechanism allows the model to weight important time steps in the sequence.

5.2 Sequence Length and Horizon

- Sequence length: 48 hours
- Forecast horizon: 1 hour

This design provides a two-day historical window while predicting short-term storm intensity.

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6. Training Strategy

6.1 Learning Rate Schedule

The model uses a schedule that alternates every 5 epochs:

- $1e-3$ for 5 epochs
- $1e-4$ for 5 epochs

This periodic schedule helps the model escape shallow minima while refining performance.

6.2 Batch Size and Epochs

- Typical batch size: 48
- 30 to 50 epochs commonly used

Checkpointing has been added so that training can be stopped and resumed without losing progress.

6.3 Loss and Metrics

- Loss: Mean Squared Error (MSE)
- Metric: Root Mean Squared Error (RMSE)

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7. Evaluation Results

7.1 Window Evaluations

Two evaluation windows were tested:

- 2023-02-14 to 2023-05-05
- 2024-04-24 to 2024-05-31

The model achieved high correlation (>0.94) and low RMSE on both windows.

7.2 Full-Year 2023 Evaluation

The full-year evaluation of 2023 produced the following results:

- RMSE: 5.117 nT
- MAE: 3.880 nT
- Correlation: 0.9586

These metrics indicate strong skill for operational Dst forecasting under observed solar wind conditions.

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8. 2025 Prediction Report (Forward Inference)

To generate predictions for 2025, solar wind inputs were available but some features were missing for that year. A mean-fill strategy was applied to missing features. This reduced accuracy relative to 2023, but still provided an operational forecast.

Summary:

- RMSE: 12.894 nT
- MAE: 10.021 nT
- Correlation: 0.7401

These results highlight the sensitivity of performance to input completeness. The model still provides useful forecasts, but better upstream data coverage would improve accuracy.

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9. Solar Wind Forecasting Model (6-hour Horizon)

To predict Dst without future solar wind measurements, we added a multi-output model that predicts the solar wind fields themselves.

9.1 Targets

The solar wind model predicts 6 hours ahead for all key wind fields:

- Magnetic field: B, Bx, By, Bz (GSE/GSM)

- Velocity: V_x , V_y , V_z , flow speed
- Plasma: density, temperature, flow pressure
- Derived: electric field, plasma beta, Alfvén Mach, magnetosonic Mach

9.2 Architecture

The model uses the same LSTM+Attention stack, but outputs a vector of all wind fields at once. This is a multi-output regression model, which captures correlations among targets.

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10. End-to-End Forecasting Pipeline

The end-to-end pipeline supports two modes:

1. Observed-mode (highest accuracy)
 - Inputs: real solar wind and IMF data
 - Output: Dst prediction 1 hour ahead
2. Forecast-mode (future years)
 - Stage 1: solar wind model predicts 6-hour future wind fields
 - Stage 2: Dst model predicts geomagnetic response from those fields

This two-stage system allows forecasts beyond the present observation window, enabling scenario planning and future-year estimates.

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11. System Design and Deployment

The project is structured into data processing, model training, evaluation, and web deployment. Key system design goals:

- Reproducible training (fixed scripts and saved metadata)
- Clear separation of datasets, models, logs
- Models stored on a high-capacity data drive
- A web front-end for visualization

The codebase also supports precomputed reports and figures for demonstrations and hackathon presentations.

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12. Limitations

1. Missing Features

- Some features are missing in late years, forcing mean-fill or drop strategies.

2. Forecast Compounding

- When solar wind is forecasted and then used for Dst, errors compound.

3. Non-stationarity

- Long-term changes in solar wind distribution can reduce model performance.

4. Limited External Inputs

- The model relies purely on in-situ solar wind measurements. It does not directly model coronal events.

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13. Future Work

- Add uncertainty estimation (quantile regression or ensembles)
- Integrate external forecast sources as additional inputs
- Train specialized storm-event models (rare extreme storms)
- Build a real-time monitoring pipeline for continuous forecasts

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14. Conclusion

This project demonstrates a complete ML pipeline for geomagnetic storm prediction. The Dst LSTM+Attention model delivers strong performance on observed data and provides a foundation for operational space weather forecasting. By adding a solar wind forecasting model, the system can predict future Dst when direct inputs are not yet available, enabling forward-looking hazard assessment.

The pipeline is deployable, extensible, and suitable for live dashboards or operational use. Its modular design also supports future integration with additional forecasting sources or physics-based models.

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15. References and Reproducibility Notes

This report is based on the codebase and datasets currently in the project workspace. All training scripts log metadata (feature list, horizon, sequence length, normalization stats) to allow reproducible runs. Model outputs and reports are saved on the data drive for audit and presentation.

Artifacts used in this report:

- Dst model: /media/ai/DATADRIIVE1/kl/models/dst_lstm_attention.keras
- 2023 report: /media/ai/DATADRIIVE1/kl/models/reports/dst_full_year_2023.md

- 2025 report: /media/ai/DATADRIVE1/kl/models/reports/dst_full_year_2025.md
- Figures: /media/ai/DATADRIVE1/kl/models/figures/

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Appendix A: Code Walkthrough (After Page 16)

This appendix explains the core code and how it maps to the paper sections.

A1. Dst Training Script

File: src/train_dst_lstm_attention.py

Key components:

- FEATURE_COLS: defines all input features for Dst prediction.
- _load_hourly_omni: loads solar wind and resamples to hourly means.
- _load_dst_csv: loads and aligns Dst labels.
- _interpolate_missing: cubic spline interpolation with a 24-hour limit.
- _build_model: defines the BiLSTM+Attention architecture.
- _lr_schedule: alternates learning rate every 5 epochs.
- train: full training pipeline, with optional resume and checkpointing.

A2. Dst Evaluation

File: src/eval_dst_windows.py

- Evaluates model on fixed windows and writes CSV/PNG outputs.
- Includes options to skip interpolation or fill missing values.

A3. Plotting Reports

File: src/plot_dst_figures.py

- Generates correlation heatmap of features
- Generates full-window plots and CSV data
- Used for 2023 full-year report and window plots

A4. Solar Wind Training

File: src/train_solar_wind_lstm.py

- Multi-output LSTM+Attention that predicts all solar wind fields 6h ahead
- Uses the same training pattern and learning rate schedule as Dst

A5. Solar Wind Forecasting

File: src/forecast_solar_wind.py

- Autoregressive forecasting in 6-hour steps
- Produces a future solar wind CSV that can feed the Dst model

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Appendix B: Key Commands

Train Dst:

```
python3 -u src/train_dst_lstm_attention.py --omni-csv /media/ai/DATADrive1/kl/data/processed/omni
```

Train Solar Wind (6h ahead, multi-output):

```
python3 -u src/train_solar_wind_lstm.py --omni-csv /media/ai/DATADrive1/kl/data/processed/omni
```

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Appendix C: ML Essentials Cheat Sheet

This section lists essential ML concepts useful for understanding and presenting the project.

C1. Data Concepts

- Train/validation/test split: separate data for fitting, tuning, and final evaluation.
- Normalization: standardizing input features improves stability and convergence.
- Missing data strategies: drop, interpolate, or fill with statistics.

C2. Model Concepts

- LSTM: recurrent architecture for sequences, good for temporal patterns.
- Attention: allows the model to focus on important time steps.
- Multi-output regression: one model predicts many targets at once.

C3. Training Concepts

- Loss: MSE is common for regression, RMSE is easier to interpret.
- Learning rate schedule: controls step size and affects convergence.
- Early stopping: prevents overfitting by halting training when val error stops improving.
- Checkpointing: saves best or latest models during training.

C4. Evaluation Concepts

- RMSE: measures average error magnitude in the same units as target.
- MAE: average absolute error; more robust to outliers than RMSE.
- Correlation: measures similarity of trends, not absolute error.

C5. Deployment Concepts

- Inference latency: how fast predictions are produced.
- Model reproducibility: saved metadata (features, normalization stats) ensures consistent results.
- Pipeline monitoring: track data quality and performance drift over time.

End of Report